

Prasanjit Dey

PhD, Computer Science

AI for Environmental Prediction · Spatiotemporal and Physics-Informed Modelling · Uncertainty Quantification

Postdoctoral Researcher, Chalmers University of Technology | ESA BIOMASS Mission

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· Google Scholar

Profile

I build machine learning systems for **environmental prediction**, and my research has consistently addressed the question that decides whether such predictions are usable: not predictive skill alone, but **how much confidence attaches to a forecast, and whether it holds where the observations are thin**. My doctorate, completed June 2026 and funded competitively by Science Foundation Ireland, developed deep architectures that fuse satellite retrievals, ground observation networks and ERA5 reanalysis to estimate and forecast near surface atmospheric fields. I am now a postdoctoral researcher at Chalmers University of Technology working on the **ESA BIOMASS mission**, developing and validating retrieval algorithms for global forest above ground biomass. This is an operational Earth observation product underpinning carbon stock assessment and climate mitigation monitoring.

Three strands of that record map onto the Environment Forecasting Mission. **Modelling complex physical and spatiotemporal systems**: hybrid architectures in which learned components are coupled to explicit transport dynamics, so predictions remain physically consistent rather than merely well fitted. **Forecasting where data is scarce**: parameter efficient adaptation of large pretrained models with demonstrated few shot and zero shot transfer into regions without dense monitoring, which is the central obstacle to operational climate intelligence for global development and adaptation. **Uncertainty quantification**: evidential and distribution free calibration, so that a forecast carries an honest interval a decision maker can act on.

25 peer reviewed outputs, 2 granted patents, and a record of carrying research through to systems that external stakeholders relied on operationally, among them a national ministry, regulators and site operators.

Education

PhD in Computer Science | Technological University Dublin, Ireland

Sep 2022 – Jun 2026

- Completed June 2026. Thesis: *Monitoring and Short-term Forecasting of Atmospheric Air Pollutants Using Deep Neural Networks*
- Supervisors: Dr. Bianca Schoen-Phelan (TU Dublin) and Dr. Soumyabrata Dev (Trinity College Dublin)
- Funded competitively by the D-REAL Centre for Research Training, Science Foundation Ireland

M.Tech in Information Technology | MAKAUT, India

2016

B.Tech in Computer Science and Engineering | WBUT, India

2013

Research Experience

Postdoctoral Researcher | Chalmers University of Technology, Gothenburg, Sweden

Sep 2026 – Present

- **ESA BIOMASS mission**, project: *Algorithm Development and Calibration/Validation of the BIOMASS Above-Ground Biomass (AGB) Product*
- Developing retrieval algorithms estimating global forest above-ground biomass from **P-band synthetic aperture radar**, the observational basis for carbon stock assessment and forest related climate mitigation monitoring
- Building processing and inference pipelines over multi temporal, multi polarisation SAR stacks: speckle handling, geometric and radiometric correction, feature extraction and retrieval
- **Calibration and validation** of the operational product against in situ and airborne reference data, with quantified uncertainty attached to retrieved estimates
- Working to agreed technical milestones within an international ESA mission consortium

Doctoral Researcher, AI for Atmospheric Prediction | Technological University Dublin,
D-REAL Centre

Sep 2022 – Jun 2026

- Self directed research programme: defined the research questions, designed the studies, built the complete software stack, ran and interpreted the experiments, and wrote and revised all publications
- Developed **physics informed and hybrid architectures** (NeuroDDAF, AirGRU) coupling graph representations of observation networks with explicit diffusion and advection dynamics and neural ODE components, returning calibrated uncertainty as a first class output
- Developed **transformer, graph and generative architectures** for multistep spatiotemporal forecasting from fused satellite, in situ and ERA5 reanalysis inputs, with explicit treatment of structured (non random) missingness such as persistent cloud obscuration

- Developed **parameter efficient adaptation of large pretrained models** (GPT4AP, One-for-All) for multivariate forecasting, evaluated on few shot and zero shot transfer into data scarce regions rather than only in distribution
- Engineered multi terabyte data pipelines with distributed and mixed precision training on HPC; released code publicly with pinned dependencies, seeded runs and container images enabling end to end reproduction of published results
- **Mentored junior doctoral researchers** daily, covering experimental design, code review, analysis and publication, with several collaborations resulting in joint papers

May – Aug 2025

Visiting Research Scientist | TU Delft, Geoscience and Remote Sensing, Netherlands

- Independently secured, funded and delivered a fixed term international research placement with defined outputs, embedded in an interdisciplinary team of geoscientists and engineers
- Led development of the coupled dynamics and learning framework; collaboration continues as a joint publication partnership

Dec 2018 – Oct 2020

Research Engineer (Project Assistant III) | CSIR Central Institute of Mining and Fuel Research, India

- Technical lead for **deployed environmental monitoring and prediction systems**: distributed sensor networks with real time inference, from requirements and design through calibration, drift compensation and field validation to operational handover
- Delivered findings to a **national ministry, safety regulators and site operators** through demonstrations, technical briefings and reports written for non specialist decision makers acting on the output
- Co-inventor on two granted Indian patents arising from the work

Dec 2020 – Aug 2022

Junior Research Fellow | NIT Jamshedpur, India

- Deep learning for environmental prediction from satellite and radar observations; curation, quality control and pipeline engineering for large geospatial and meteorological datasets

Feb 2017 – Jul 2018

Assistant Professor | Ramgovind Institute of Technology, Koderma, India

- Sole lecturer for four undergraduate Computer Science courses, built from nothing; supervised six final year research projects to completion and coordinated project allocation across the cohort

Technical Expertise

Spatiotemporal modelling	Physics informed and transport constrained architectures, neural ODEs, graph neural networks over observation networks, multistep forecasting, learned super resolution and downscaling, structured missing data methods
Uncertainty quantification	Evidential deep learning, distribution free and conformal calibration, aleatoric/epistemic decomposition, out of distribution and drift detection, validation against reference observations
Machine learning	Vision transformers, CNNs, variational and generative models, foundation model and parameter efficient adaptation (LoRA/PEFT), few shot and zero shot transfer, model compression and quantisation
Environmental data	ERA5 reanalysis, Sentinel and MODIS retrievals, P-band SAR, ground observation networks, low cost sensor arrays; NetCDF, HDF5, GeoTIFF, Zarr, CF conventions, xarray
Programming	Python (expert; PyTorch, TensorFlow, NumPy, SciPy, pandas, xarray, scikit-learn), C, MATLAB, R (working), SQL, Bash
Computing & engineering	Linux, HPC and job scheduling, multi node and mixed precision GPU training, CUDA (NVIDIA certified), Git/GitHub, Docker, CI, testing, reproducible pipelines
Open research	Public ORCID, openly licensed preprints, code released with pinned dependencies, seeded runs and container images; FAIR data practice with documented metadata and provenance

Research Leadership, Mentoring and Collaboration

- **Mentoring**: daily mentoring of junior doctoral researchers (TU Dublin, 2023–2026) in experimental design, code review, analysis and publication, with joint authorship on the resulting papers
- **Supervision**: six undergraduate research projects supervised to completion; taught masters project students in applied machine learning and environmental data analysis
- **Independent project delivery**: secured, funded, planned and delivered an international research placement

- (TU Delft, 2025) end to end; delivered field deployed systems to fixed milestones with external accountability to a government ministry (CSIR-CIMFR)
- **Shaping research direction:** defined my own doctoral research programme and its successive research questions; currently contributing to algorithm development within an ESA mission consortium
 - **International collaboration:** sustained joint authorship across Ireland, Sweden, the Netherlands, India and Singapore
 - **Peer review:** reviewer for international journals and conferences in remote sensing, environmental modelling and applied machine learning

Funding and Awards

- **D-REAL Centre for Research Training doctoral award**, Science Foundation Ireland: a competitive national award covering stipend, fees and research costs (2022–2026)
- **TU Delft visiting research appointment**, secured and organised independently, including funding and deliverables (2025)
- **CSIR-CIMFR Dr. Adinath Lahiri Award** for outstanding research publication (2021)
- **NVIDIA certification:** Fundamentals of Accelerated Computing with CUDA C/C++ (2025)
- Contributor to proposal preparation targeting national and European instruments (Research Ireland, Horizon Europe, MSCA)

Dissemination and Stakeholder Communication

- **Conference presentation:** IGARSS 2023, 2024 and 2025; iEMSs 2026 (Dublin); PIERS 2023; EI² 2023; CISP-BMEI 2024
- **Research seminars:** TU Dublin and TU Delft, including working sessions with an interdisciplinary group of geoscientists and engineers
- **Non specialist and policy facing communication:** technical briefings, demonstrations and written reports for ministry officials, safety regulators and operational staff whose decisions depended directly on model output
- **Teaching:** four undergraduate courses as sole lecturer; laboratory and tutorial teaching in scientific computing, data analysis, software engineering and robotics

Complete List of Publications

All outputs registered on my public ORCID record: <https://orcid.org/0000-0002-2248-1701>. Totals: **25 peer reviewed outputs**, comprising 14 journal articles (10 published, 4 under review) and 11 refereed conference papers, plus **2 granted patents** and open code releases. Openly licensed preprints available for all recent manuscripts.

Journal Articles, Published (10)

1. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2026). PollutionNet: A Vision Transformer Framework for Climatological Assessment of NO₂ and SO₂ Using Satellite-Ground Data Fusion. *Theoretical and Applied Climatology*, 157, 509. doi:10.1007/s00704-026-06376-8
2. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2024). CombineDeepNet: A Deep Network for Multistep Prediction of Near-Surface PM_{2.5} Concentration. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17, 788–807. doi:10.1109/JSTARS.2023.3333269
3. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2024). Predicting Multivariate Air Pollution: A Gaussian-Mixture Nested Factorial Variational Autoencoder Approach. *IEEE Geoscience and Remote Sensing Letters*, 21, 1–5. doi:10.1109/LGRS.2024.3416343
4. **Dey, P.**, Das, B.P., Lee, Y.H., Dev, S. (2023). NeSNet: A Deep Network for Estimating Near-Surface Pollutant Concentrations. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 3797–3804. doi:10.1109/JSTARS.2023.3244719
5. **Dey, P.**, Saurabh, K., Kumar, C., Pandit, D., Chaulya, S.K., Ray, S.K., Prasad, G.M., Mandal, S.K. (2021). t-SNE and variational auto-encoder with a bi-LSTM neural network-based model for prediction of gas concentration in underground coal mines. *Soft Computing*, 25(22), 14183–14207. doi:10.1007/s00500-021-06261-8
6. **Dey, P.**, Chaulya, S.K., Kumar, S. (2021). Hybrid CNN–LSTM and IoT-based coal mine hazards monitoring and prediction system. *Process Safety and Environmental Protection*, 152, 249–263. doi:10.1016/j.psep.2021.06.005
7. **Dey, P.**, Chaulya, S.K. (2021). Secure decision tree twin support vector machine training and classification for encrypted IoT data via blockchain platform. *Concurrency and Computation: Practice and Experience*. doi:10.1002/cpe.6264
8. Kumari, K., **Dey, P.**, Kumar, C., Pandit, D., Mishra, S.S., Kisku, V., Chaulya, S.K., Ray, S.K., Prasad, G.M. (2021). UMAP and LSTM based fire status and explosibility prediction for sealed-off area in underground coal mine. *Process Safety and Environmental Protection*, 146, 837–852. doi:10.1016/j.psep.2020.12.019
9. **Dey, P.**, Kumar, C., Mitra, M., Mishra, R., Chaulya, S.K., Prasad, G.M., Mandal, S.K., Banerjee, G. (2021).

Deep convolutional neural network based secure wireless voice communication for underground mines. *Journal of Ambient Intelligence and Humanized Computing*. doi:10.1007/s12652-020-02700-w

10. Hati, S., **Dey, P.**, De, D. (2017). WLAN based energy efficient smart city design. *Microsystem Technologies*, 25(5), 1599–1612. doi:10.1007/s00542-017-3530-6

Journal Articles, Under Review (4)

11. **Dey, P.**, Dev, S., Mayer, A., Schoen-Phelan, B. (2026). NeuroDDAF: Neural Dynamic Diffusion-Advection Fields with Evidential Fusion for Air Quality Forecasting. *arXiv:2604.01175*. Under review. Code released publicly.
12. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2026). Meteorology-Driven GPT4AP: A Multi-Task Forecasting LLM for Atmospheric Air Pollution in Data-Scarce Settings. *arXiv:2603.29974*. Under review. Code released publicly.
13. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2026). One-for-All: A Lightweight Stabilized and Parameter-Efficient Pre-trained LLM for Time Series Forecasting. *arXiv:2603.29756*. Under review.
14. **Dey, P.**, Yahn, Z., Dev, S., Schoen-Phelan, B. (2026). TinyNina: A Resource-Efficient Edge-AI Framework for Sustainable Air Quality Monitoring via Intra-Image Satellite Super-Resolution. *arXiv:2604.04445*. Under review.

Refereed Conference Papers (11)

1. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2026). AirGRU: A Physics-Informed Model for PM_{2.5} Prediction. *Proceedings of the 13th International Congress on Environmental Modelling and Software (iEMSs 2026)*, Dublin, Ireland.
2. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2025). Efficient Vision Transformer Framework: Near-Surface O₃ Prediction with Missing Satellite Data. *IGARSS 2025*. IEEE. doi:10.1109/igarss55030.2025.11244019
3. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2025). FewShot-Airnet: Temporal 2D-Variation Modeling for Air Pollution Forecasting. *IGARSS 2025*. IEEE. doi:10.1109/igarss55030.2025.11243753
4. Kumari, S., **Dey, P.**, Ayala-Cabrera, D., Dev, S. (2024). Attention-Based ResNet for Accurate Water Body Classification Using Spectral Indices. *CISP-BMEI 2024*. IEEE. doi:10.1109/cisp-bmei64163.2024.10906094
5. **Dey, P.**, Dhang, S., Dev, S., Schoen-Phelan, B. (2024). NES-VIT-NET: A Nested Vision Transformer-Based Network for Near-Surface NO₂ Estimation. *IGARSS 2024*. IEEE. doi:10.1109/igarss53475.2024.10640856
6. Dhang, S., **Dey, P.**, Zhang, M., Dev, S. (2024). Use-Net: Satellite Data-Based Framework for Optimizing Billboard Placement in Urban Areas. *IGARSS 2024*. IEEE. doi:10.1109/igarss53475.2024.10641669
7. Akrami, N., Li, Y., **Dey, P.**, Dev, S. (2023). Spatial–Temporal–TES: Reanalysis Dataset based Short-Term Temperature Forecasting System. *EI2 2023*. IEEE. doi:10.1109/ei259745.2023.10512739
8. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2023). BiLSTM–BiGRU: A Fusion Deep Neural Network For Predicting Air Pollutant Concentration. *IGARSS 2023*. IEEE. doi:10.1109/igarss52108.2023.10282742
9. Abdel Azeem, M., **Dey, P.**, Dev, S. (2023). A Multidimensionality Reduction Approach to Rainfall Prediction. *PIERS 2023*. IEEE. doi:10.1109/piers59004.2023.10221498
10. **Dey, P.**, Dev, S., Schoen-Phelan, B. (2023). NeSDeepNet: A Fusion Framework for Multi-step Forecasting of Near-surface Air Pollutants. *PIERS 2023*. IEEE. doi:10.1109/piers59004.2023.10221327
11. Agarwal, A., **Dey, P.**, Kumar, S. (2022). Sentiment Analysis using Modified GRU. *Contemporary Computing 2022*. ACM. doi:10.1145/3549206.3549270

Granted Patents (2)

1. **Dey, P.**, De, D., Hati, S. (2023). *Location Tracking System for Indoor Environment (LTIE)*. Indian Granted Patent No. 465850. Application No.: 201831030620. Filed: 16 August 2018. Granted: 06 November 2023.
2. Chaulya, S.K., . . . , **Dey, P.**, et al. (2026). *Digital Mine Using Internet of Things*. CSIR India. Indian Granted Patent No. 580680. Application No.: 202011029344. Filed: 10 July 2020. Granted: 16 February 2026.

Open Software and Data Releases

- **NeuroDDAF**, a physics informed framework with evidential uncertainty fusion.
github.com/Prasanjit-Dey/NeuroDDAF
- **One_for_All**, parameter efficient adaptation of pretrained models for time series forecasting.
github.com/Prasanjit-Dey/One_for_All
- All releases include pinned dependencies, configuration files, seeded random states and container images enabling end-to-end reproduction of published results. Derived datasets and preprocessing pipelines are provided in community-standard formats with documented provenance.

Referees

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